Indian Institute of Technology Bombay
MA 401 Linear Algebra

Mid-Sem Exam

Date: September 18, 2024 | Maximum Marks: 34

Duration: 2 hours

Note:

1. Use results which are covered in the class only. $\mathbb{R}$ is the set of all real numbers.
2. Justify all of your answers. Answer without proper justification will be treated as no answer.

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1. Let $\varphi : M_{n\times n}(\mathbb{R}) \to \mathbb{R}$ be the linear functional defined by $\varphi(A) = \sum_{i,j=1}^{n} a_{ij}$ if $A = (a_{ij})_{i,j=1}^{n}$. Let $\psi : M_{n\times n}(\mathbb{R}) \to \mathbb{R}$ be the linear functional defined by $\psi(A) = \mathrm{tr}(A)$, where tr denotes the trace of a matrix.

   a) If $T : M_{n\times n}(\mathbb{R}) \to M_{n\times n}(\mathbb{R})$ is defined by $T(A) = A^T$, where A^T is the transpose of the matrix A, and if T' is the transpose map of T, then determine $T'(\varphi)$.

   b) If $T : M_{n\times n}(\mathbb{R}) \to M_{n\times n}(\mathbb{R})$ is defined by $T(A) = AB - BA$ for all $A \in M_{n\times n}(\mathbb{R})$ and for a fixed $B \in M_{n\times n}(\mathbb{R})$, and if T' is the transpose map of T, then determine $T'(\psi)$.

   [4+ 4 = 8 Marks]

2. Let A and B be $n \times n$ real matrices with $AB = 0$. Give a proof or counterexample for each of the following.

   a) Either $A = 0$ or $B = 0$.

   b) BA has trace 0.

   c) If A is similar to a matrix C with $\det(C) = -2$, then $B = 0$.

   d) There is a non-zero vector $x \in \mathbb{R}^n$ such that $BAx = 0$.

   [2+2+2+2= 8 Marks]

P.T.O.

3. Let $A \in M_{3\times 4}(\mathbb{R})$, and let $A : \mathbb{R}^4 \to \mathbb{R}^3$ be the corresponding linear map.

a) If the last two columns of A are same, show that A is not one-to-one by finding a non-zero vector in $\mathbb{R}^4$ that is in the nullspace of A.

b) If a_1, a_2, a_3, and a_4 are the columns of A such that $a_1 + 3a_2 - 4a_3 + 2a_4 = 0$, find a non-zero vector in the nullspace of A.

c) If the first two rows of A are same, show that A is not onto by finding a vector $y \in \mathbb{R}^3$ that is not in the image of A (justify your answer).

d) Suppose b_1, b_2, b_3 are the rows of A. If $b_1 - 2b_2 + b_3 = 0$, find a vector in $\mathbb{R}^3$ which is not in the image of A (justify your answer).

[2+2+4+4= 12 Marks]

4. Let X be the vector space of polynomials of degree strictly less than n with real coefficients. Let $I_1, \ldots, I_n$ be pairwise disjoint intervals of $\mathbb{R}$ such that $0 < |I_j| < \infty$ for each $j = 1, \ldots, n$, where $|I_j|$ is the length of the interval I_j. For any $p \in X$ and $1 \leq j \leq n$, define

$$p_j := \frac{1}{|I_j|} \int_{I_j} p(t)\, dt,$$

a) Show that $T : X \to \mathbb{R}^n$ defined by $T(p) = (p_1, p_2, \ldots, p_n)$ is a linear map.

b) Find the null space N_T and the range space R_T of T.

[2+ 4= 6 Marks]

$\cos t + i \sin t$ $e^{\frac{i\pi}{2}}$ $\cos\pi + i$ $\frac{-b \pm \sqrt{\ }}{2}$ $P^{-1}BP$ λ $BAv = ABv$

Indian Institute of Technology Bombay

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QUIZ II

Date: October 23, 2024 Maximum Marks: 15

Duration: 45 minutes

Note:

1. Use results which are covered in the class only. $\mathbb{R}$ is the set of all real numbers.
2. Justify all of your answers. Answer without proper justification will be treated as no answer.

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1. Let A and B be commuting $n \times n$ matrices over $\mathbb{C}$ (i.e., $AB = BA$) and suppose that A has an eigenvector v with eigenvalue λ.

   (a) Show that if $Bv \neq 0$ then Bv is also an eigenvector of A with eigenvalue λ.

   (b) If eigenvalues of A are all distinct, what is the dimension of each eigenspace of A?

   (c) If eigenvalues of A are all distinct, show that there exists an invertible matrix P such that both $P^{-1}AP$ and $P^{-1}BP$ are diagonal.

   [2+2+3= 7 Marks]

2. (a) Let $t \in \mathbb{R}$ be such that t is not an integer multiple of π. Let

$$A = \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}.$$

   Show that there does not exist a real valued invertible matrix P such that $P^{-1}AP$ is a diagonal matrix.

   (b) Determine all $\lambda \in \mathbb{R}$ for which the matrix A is not diagonalizable over $\mathbb{R}$, where

$$A = \begin{bmatrix} 1 & \lambda \\ 0 & 1 \end{bmatrix}.$$

   [3+2=5 Marks]

3. Let $A \in M_{n \times n}(\mathbb{C})$ be such that $A^m = I$ for some positive integer m. Is A diagonalizable over $\mathbb{C}$ (Justify your answer)?

   [3 Marks]

$x^3 = 1$ $x = e$ Yes. $Av = \lambda v$. $x^m = 1$ distinct roots $AB = BA$ $A^m = I$ $x^m - 1$ $Bv \neq 0$ $Av = \lambda v$ $BAv = B\lambda v$ $ABv = \lambda Bv$ ABv $\boxed{A^m = I}$ $\cos^2 t + \sin^2 t = 1$ $\sin^2 t = 1 - \cos^2 t$ $-\sin^2 t$

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MA 401 Linear Algebra

End-Sem Exam

Date: November 13, 2024 Maximum Marks: 40
Duration: 2 hours

Note:

1. Use results which are covered in the class only. $\mathbb{R}$ denotes the set of all real numbers and $M_{n\times n}(\mathbb{R})$ denotes the set of all $n \times n$ real matrices.

2. Justify all of your answers. Answer without proper justification will be treated as no answer.

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1. Let $A \in M_{4\times 4}(\mathbb{R})$ be such that the eigenvalues of A are only 1 and -1. Find the determinant of $A^{1000} - A^{100} + A^{10} - I$. $= 0$ [3 Marks]

2. Let R, L be linear maps on $\mathbb{C}^n$ defined by

$$R(x_1, x_2, \ldots, x_n) = (0, x_1, x_2, \ldots, x_{n-1}) \text{ and } L(x_1, x_2, \ldots, x_n) = (x_2, x_3, \ldots, x_n, 0),$$

for all $(x_1, x_2, \ldots, x_n) \in \mathbb{C}^n$. Find matrix representations of R and L. Are R and L similar (Justify your answer)? [2 + 1= 3 Marks]

3. Let $\mathbb{R}^3$ be the inner product space with usual inner product. Let $S = \text{Span}\{(1, -1, 0), (1, -1, 1)\}$. Let P and R be the orthogonal projections onto S and $S^{\perp}$ respectively.
(0,0,1) a) Find an orthonormal basis for S and use it to find $P(1, 1, 1)$. $= (0,0,1)$
b) Find an orthonormal basis for $S^{\perp}$ and use it to find $R(1, 1, 1)$. [3+2= 5 Marks]

4. If A is a matrix in $M_{5\times 5}(\mathbb{C})$ with characteristic polynomial $(x-2)^3(x+5)^2$ and minimal polynomial $(x-2)^2(x+5)$, what is the Jordan form of A? [3 Marks]

5. Find a real 2×2 matrix A, other than $A = I$ such that $A^5 = I$. [4 Marks]

P.T.O.

6. For each of the following, answer TRUE or FALSE. If the answer is TRUE then provide a proof. Otherwise, provide a counterexample.

a) The set $\{A \in M_{n \times n}(\mathbb{R}) : 2A - A^T = 0\}$ is a subspace of $M_{n \times n}(\mathbb{R})$, where A^T is the transpose of A.

b) The set $\{A \in M_{3\times3}(\mathbb{R}) : A(e_1) = (0,0,0)\}$ is a subspace of $M_{3\times3}(\mathbb{R})$, where $e_1 = (1,0,0)$.

c) If $A \in M_{2\times3}(\mathbb{R})$ and $B \in M_{3\times2}(\mathbb{R})$, then BA can not be invertible.

d) If $A \in M_{2\times3}(\mathbb{R})$ and $B \in M_{3\times2}(\mathbb{R})$, then AB can not be invertible.

e) If $A \in M_{2\times2}(\mathbb{R})$ and if $A^2 = A$ then either $A = 0$ or $A = I$.

f) If $A \in M_{n \times n}(\mathbb{C})$ and if $A - 2I = -A^2$, then A is invertible.

[8 Marks]

7. Let $A = (a_{ij}) \in M_{n \times n}(\mathbb{C})$ $(n > 1)$ be such that rank of A is 1. Let $v := (v_1, ..., v_n) \neq 0$ be a basis for the image of A.

(a) Show that for all $i, j = 1, \ldots, n$, $a_{ij} = v_i w_j$ for some vector $w = (w_1, \ldots, w_n) \neq 0$.

(b) How the eigensapce of A corresponding to the eigenvalue 0 is related to w?

3

[5+3 = 8 Marks]

8. Let $A \in M_{3\times3}(\mathbb{C})$, and let $S \in M_{3\times3}(\mathbb{C})$ be invertible. If $SA = -AS$, show that 0 is an eigenvalue of A.

[3 Marks]

9. Let $T \in M_{n \times n}(\mathbb{C})$ be a self-adjoint matrix. If trace of A^2 is zero, show that $A = 0$.

[3 Marks]

$A^* = A$

$AA^* = A^2$

$tr(AA^*) = tr\ 0$

$\sum^n \sum^n a_{ij}\overline{a_{ij}} = 0$